Aram Ansary Ogholbake

Department of Computer Science and Department of Internal Medicine
University of Kentucky
Lexington, KY 40506

Email: aram.ansary@uky.edu

Research Area

- Deep Learning, Machine Learning, Omics Data, Circadian Rhythms, Biomedical Informatics

Education

Ph.D. in Computer Science, University of Kentucky

Expected Graduation Date: May 2026
GPA: 3.87

Master of Science in Finance, University of Kentucky

Graduation Date: May 2020
GPA: 3.7

Bachelor of Science in Mathematics, University of Kentucky

Graduation Date: May 2018
GPA: 3.71

Publications

Published

Ansary Ogholbake, A., & Cheng, Q. (2025). Five-Gene Expression Formula Accurately Detects Hepatocellular Carcinoma Tumors. Biotechnology Journal, 20(7), e70073.

Ansary Ogholbake, A., & Cheng, Q. (2025). PROTECT: Protein circadian time prediction using unsupervised learning. iScience, 28(10).

Ansary Ogholbake, A., & Cheng, Q. (2024). Gene expression clock: an unsupervised deep learning approach for predicting circadian rhythmicity from whole genome expression. Neural Computing and Applications, 36(33), 20653-20670.

Ansary Ogholbake, A., & Cheng, Q. (2023). PENN: Phase Estimation Neural Network on Gene Expression Data. In *The International Conference on Deep Learning, Big Data and Blockchain* (pp. 59-67). Cham: Springer Nature Switzerland.

Archbold, C., Brodie, B., Ansary Ogholbake, A., & Jacobs, N. (2023). Fine-Grained Property Value Assessment Using Probabilistic Disaggregation. In *IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium* (pp. 5312-5315). IEEE.

Ansary Ogholbake, A., & Khamfroush, H. (2021). A modified seir model for the spread of covid-19 considering different vaccine types. *arXiv preprint arXiv:2106.08820*.

In Review

A minimal three-gene molecular signature enables precise early lung adenocarcinoma detection.

Symbolic cross-platform diagnostic formula from bulk to single-cell transcriptomics using Kolmogorov--Arnold networks

PATNet: Specialized MLP Architectures Outperform Large Language Models in Patent Approval Time Prediction

In Prep

A Multimodal Deep Learning Framework for Edema Classification Using HCT and Clinical Data

The Anatomy of Causal Learning in Biomedicine: From Statistical Estimation to Reasoning

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Patents

Ansary Ogholbake, A., Cheng, Q. Hepatocellular Carcinoma Assessment Using Peripheral Blood Mononuclear Cells. U.S. Provisional Patent Application No. 63/842,414, filed July 11, 2025.

Honors and Awards

- Computer Science Department Outstanding Student Paper award (2025)
- Master of Science in Finance Scholarship
- Member of 'The National Society of Collegiate Scholars (NSCS)'
- Dean's List

Teaching Activities

- Spring 2023: Full Teaching Assistantship, University of Kentucky, Department of Computer Science
- Fall 2021: Full Teaching Assistantship, University of Kentucky, Department of Computer Science
- Spring 2021: Grader, University of Kentucky, Department of Computer Science
- Spring 2020: Full Teaching Assistantship, University of Kentucky, Department of Physics & Astronomy
- Spring 2019: Paid Tutor, University of Kentucky

Research and Scientific Experience

Spring 2023-Present

Research under supervision of Dr. Cheng at the Computer Science Department, University of Kentucky
Research in artificial intelligence and machine learning methods for analyzing complex biomedical data, including omics, medical imaging, and clinical datasets, with applications to disease modeling and biomedical discovery.

Spring 2021-Fall 2022

Research under supervision of Dr. Jacobs at the Computer Science Department, University of Kentucky
Research in Image Segmentation, Change Detection, Regression on Weak Labels, Single Image Depth Estimation

Spring 2020

Worked on Algebraic Number Theory as an independent study with Dr. David Leep at the Mathematics Department of the University of Kentucky

Spring 2019

Worked on Quadratic Reciprocity as an independent study with Dr. David Leep at the Mathematics Department of the University of Kentucky

Technical Skills

Programming

Python, C++, C, R, Julia

Libraries

PyTorch Lightning, TensorFlow, Keras, OpenCV, sumolib